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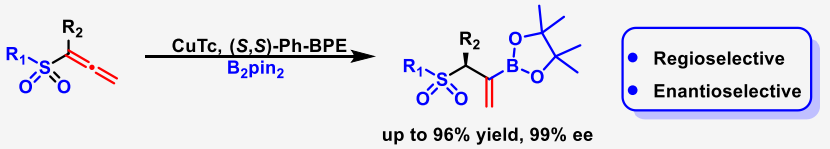
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Copper-Catalyzed Regio- and Enantioselective Protoboration of Allenyl Sulfones to Access Chiral Allylic Sulfones

Peidong Sun,† Xiaomei Kong,† Jingwen Huang, Xuedi Ma, Tongyu Xu,* and Fanlong Zeng*

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ABSTRACT: An efficient method for copper-catalyzed asymmetric protoboration of allenyl sulfones with bis(pinacolato)diboron was developed, providing chiral allylic sulfones with high efficiency and excellent enantioselectivity. Notably, the directing effect of the sulfone group plays a pivotal role in achieving both high regioselectivity and enantioselectivity.

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INTRODUCTION

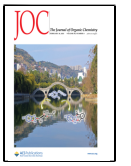
The development of efficient and reliable methods for achieving regio- and enantioselective synthesis of chiral compounds has long been a significant objective in organic synthesis. Chiral sulfones serve as versatile building blocks in organic chemistry and are crucial motifs in materials and biologically active molecules.[1−3] Recently, the transition metal-catalyzed preparation of chiral allylic sulfones has garnered increasing attention, leading to the advancement of various asymmetric strategies. For instance, enantioselective allylic substitution has been established to yield α-monosubstituted and α,α-disubstituted chiral allylic sulfones from diverse allylic precursors and sulfinyl nucleophiles in the presence of Pd,[4] Rh,[5] Ir,[6] Fe,[7] and W[8] catalysts. Additionally, chiral allylic sulfones have also been synthesized from allenes, alkynes, or dienes through asymmetric sulfonylation[9] and/or thiolation[10] followed by oxidation processes developed by different research groups. The rhodium-catalyzed asymmetric hydrogenation of allenyl sulfones offers an atom-economical approach to producing chiral allylic sulfones.[11] Furthermore, chiral sulfones can be synthesized through enantioselective photocatalytic hydrosulfonylation of α,β-unsaturated carbonyl compounds and styrenes.[12]

Despite recent successful examples, there remains a significant demand for the development of facile and cost−effective transition metal catalytic systems capable of efficiently producing chiral allylic sulfones from readily available starting materials. Previously, we reported an effective method for copper-catalyzed asymmetric protoboration of phosphinylallenes,[13] which afforded chiral allylphosphine oxides in high yields and excellent enantioselectivities (Scheme 1e). When this catalytic strategy was applied to allenyl sulfone, it resulted in the formation of chiral allylic sulfones with high enantioselectivity. In this study, we extensively investigated the efficient regio- and enantioselective protoboration of allenyl sulfones to generate chiral allylic sulfones.

RESULTS AND DISCUSSION

The regio- and enantioselective protoboration of allenyl sulfone (**1a**) with B_2pin_2 was successfully achieved, yielding chiral allylic sulfone **2a** in 70% yield and 95% enantiomeric excess (ee) in the presence of CuOAc/(*S*,*S*)-Ph-BPE. The reaction was conducted using toluene as the solvent, NaO^{t}Bu as the base, and MeOH as the proton source at −55 °C (Table 1, entry 1). The reaction temperature plays a crucial role in determining the enantioselectivity for α-carbons adjacent to electron-withdrawing groups, which may lead to racemization under basic conditions according to our previous studies. When the reaction temperature was increased to −30 °C for the protoboration of allenyl sulfone **1a**, there was no reduction in the enantioselectivity of product **2a** (entries 2 and 3). Further investigations of the asymmetric protoboration at temperatures of −20 °C or −35 °C indicated that a temperature of −35 °C is preferable, resulting in an improved yield of 80% and an ee of 98% (entries 4 and 5). Additionally, when CuOAc was replaced with various copper salts including $Cu(OAc)_2$, $Cu(CH_3CN)_4PF_6$, CuTc, CuCl, CuI, and $CuCl_2$ (entries 6−11), it was found that using CuTc as a precatalyst enhanced the reaction outcomes, affording product **2a** in 88% and 98% ee. However, lower yields were observed when reactions were performed at other temperatures ranging from −40 to −20 °C (entries 12−14). Other chiral ligands, bases,

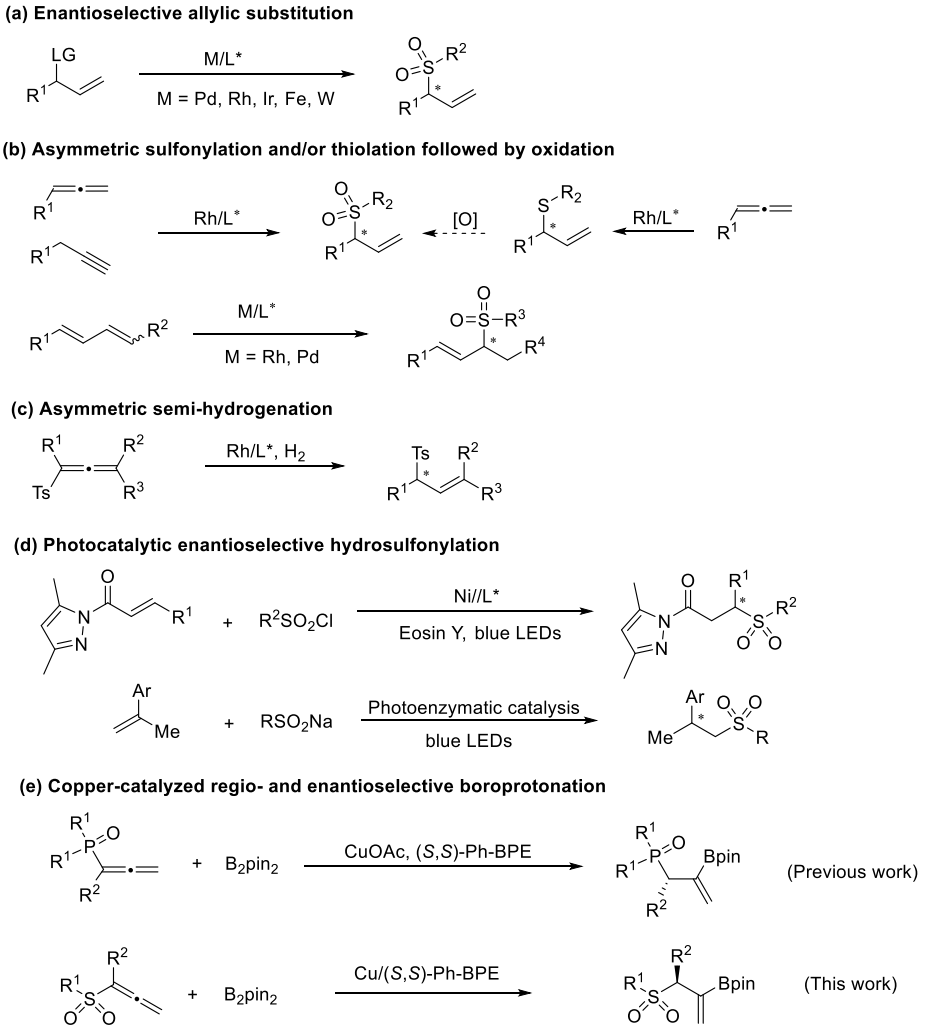
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Scheme 1. Different Strategies to Synthesize Chiral Allylic Sulfones



and solvents were also investigated; however, these modifications did not further enhance the yields or enantioselectivity of product **2a** (for details, see Tables S1−S3).

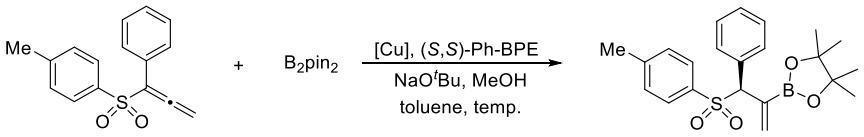
The scope of asymmetric protoboration of allenyl sulfones **1** with α-substituent was investigated in the presence CuTc/(*S*,*S*)-Ph-BPE at −35 °C in toluene (Scheme 2). Allenyl sulfones with varying *para*-, *meta*- or *ortho*-substituents on their α-phenyl moiety were evaluated, affording the corresponding chiral allylic sulfones in moderate to excellent yields with high enantioselectivities (**2b**−**l**). Asymmetric protoboration using substrates **1c**−**e** with the *para*-halogen substituents (F, Cl, and Br) proceeded smoothly, affording products **2c**−**e** in 58−68% yields with 96−98% ees. Both electron-donating and electron-withdrawing substituted substrates worked well, producing corresponding products (**2b**, **2f**−**l**) in good yields exceeding 92% ees. Furthermore, allenyl sulfones containing naphthyl substituents provided compounds **2m** and **2n** in 88% and 92% yields with 98% and 93% ee, respectively. The thienyl-substituted allenyl sulfone afforded corresponding products **2o** in 96% yield and 94% ee. Additionally, substrates having alkyl substituents such as Me, *n*-Pr, Bn and allyl resulted in good yields for compounds **2p**−**s** while maintaining ees between 96% and 98%. A gram scale reaction of **1a** was conducted, affording **2a** in 85% yield and 95% ee, thereby demonstrating the efficiency of this method.

Different allenyl sulfones were subsequently evaluated under the optimized reaction conditions (Scheme 3). The aryl sulfones bearing electron-donating and electron-withdrawing groups afforded corresponding products **4a**−**f** in good yield and high enantiomeric excesses (96−98% ee). Substituent at the *ortho*-, *meta*- or *para*-positions had minimal impact on the enantioselectivity. Heteroaryl sulfones demonstrated good reactivity, producing corresponding **4g** and **4h** in 81% and 71% yields with 98% ee values, respectively. Alkyl sulfones (*n*-Bu and Bn) underwent asymmetric protoboration smoothly, affording **4i** and **4j** in 74% and 56% yields with 99% and 97% ee, respectively. The internal allenyl sulfone **3k** was synthesized and subjected to the optimized reaction conditions; however, the desired product **4k** failed to form at both −35 °C and room temperature, with the majority of 3k remaining unreacted. In contrast, the allenyl sulfone **3l**, lacking an α-substituent, yielded compound **4l** in 50% yield. The configuration of compound **4c** was determined through X-ray crystallographic analysis, confirming the formation of the (*R*)-isomer during the reaction (CCDC 2361361; see Figure S1).

The transformations of **2a** were investigated. The halogenation products **2aa** and **2ab** were synthesized smoothly through the reaction of **2a** with the corresponding copper(II) halides (eq 1). Additionally, compound **2a** could be readily

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Table 1. Optimization of the Reaction Conditions[a]

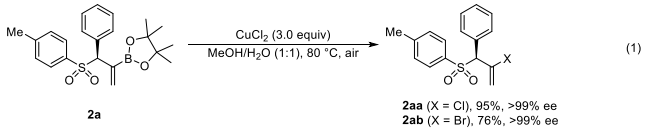


1a 2a

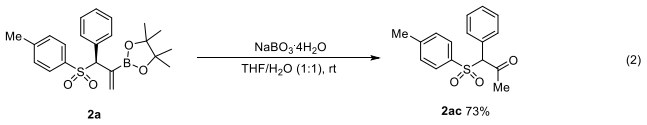
entry	[Cu]	temp. (°C)	yield (%)	ee (%)
1	CuOAc	−55	70	97
2	CuOAc	−40	76	98
3	CuOAc	−30	74	98
4	CuOAc	−20	63	96
5	CuOAc	−35	80	98
6	$Cu(OAc)_2$	−35	77	81
7	$Cu(CH_3CN)_4PF_6$	−35	62	94
8	**CuTc**	**−35**	**88**	**98**
9	CuCl	−35	67	95
10	CuI	−35	60	92
11	$CuCl_2$	−35	47	91
12	CuTc	−40	80	98
13	CuTc	−30	84	98
14	CuTc	−20	77	98

[a]Conditions: **1a** (0.2 mmol), B_2pin_2 (0.24 mmol), Cu salt (5 mol %), (*S*,*S*)-Ph-BPE (6 mol %), NaO^{t}Bu (20 mol %), MeOH (0.4 mmol), toluene (2 mL), 12 h. Isolated yields and ee values were determined by chiral HPLC analysis. CuTc = copper(I) thiophene-2-carboxylate, (*S*,*S*)-Ph-BPE = 1,2-bis((2*S*,5*S*)-2,5-diphenylphospholano)ethane.

oxidized to a ketone **2ac** using sodium perborate as the oxidizing agent (eq 2).

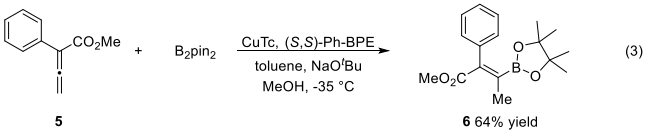


(1)



(2)

To elucidate the role of the sulfone group, allenyl ester **5** was subjected to asymmetric protoboration in the presence CuTc/(*S*,*S*)-Ph-BPE at −35 °C in toluene (eq 3). However,



(3)

compound **6** was obtained in 64% yield, while no chiral allylic ester was produced. Lin and co-workers reported on copper-catalyzed enantioselective hydroboration of 3-aryl substituted cyclopropene-3-carboxylates, where a nondirecting effect of the ester group was observed, indicating that the carboxylate does not exhibit any coordination effect with copper.[14] Based on our experimental findings and previous literature reports,[15] a plausible reaction mechanism is proposed in Scheme 4. Copper(I) alkoxide **A** is generated in situ by base treatment. Subsequently, copper-boryl complex **B** is formed via transmetalation between **A** and B_2pin_2. Due to the electron-withdrawing properties and metal coordination capabilities of the sulfone group, it is reasonable to hypothesize that borocupration of allenyl sulfone **1** with **B** may proceed through six-membered transition state **C** to form **D**, which subsequently yields chiral allylic sulfone **2** via protonolysis. In the process, both electron-withdrawing characteristics and coordination effect of sulfone groups play crucial roles in determining regio- and enantioselectivity.

CONCLUSION

In summary, an efficient protocol has been developed for the copper-catalyzed asymmetric protoboration of allenyl sulfones, affording chiral allylic sulfones in high yield with excellent enantioselectivity. Notably, the directing effect of the sulfone group is crucial for achieving both high regioselectivity and enantioselectivity.

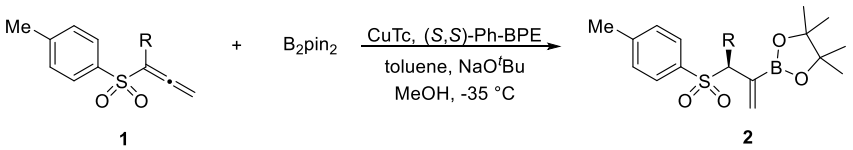
EXPERIMENTAL SECTION

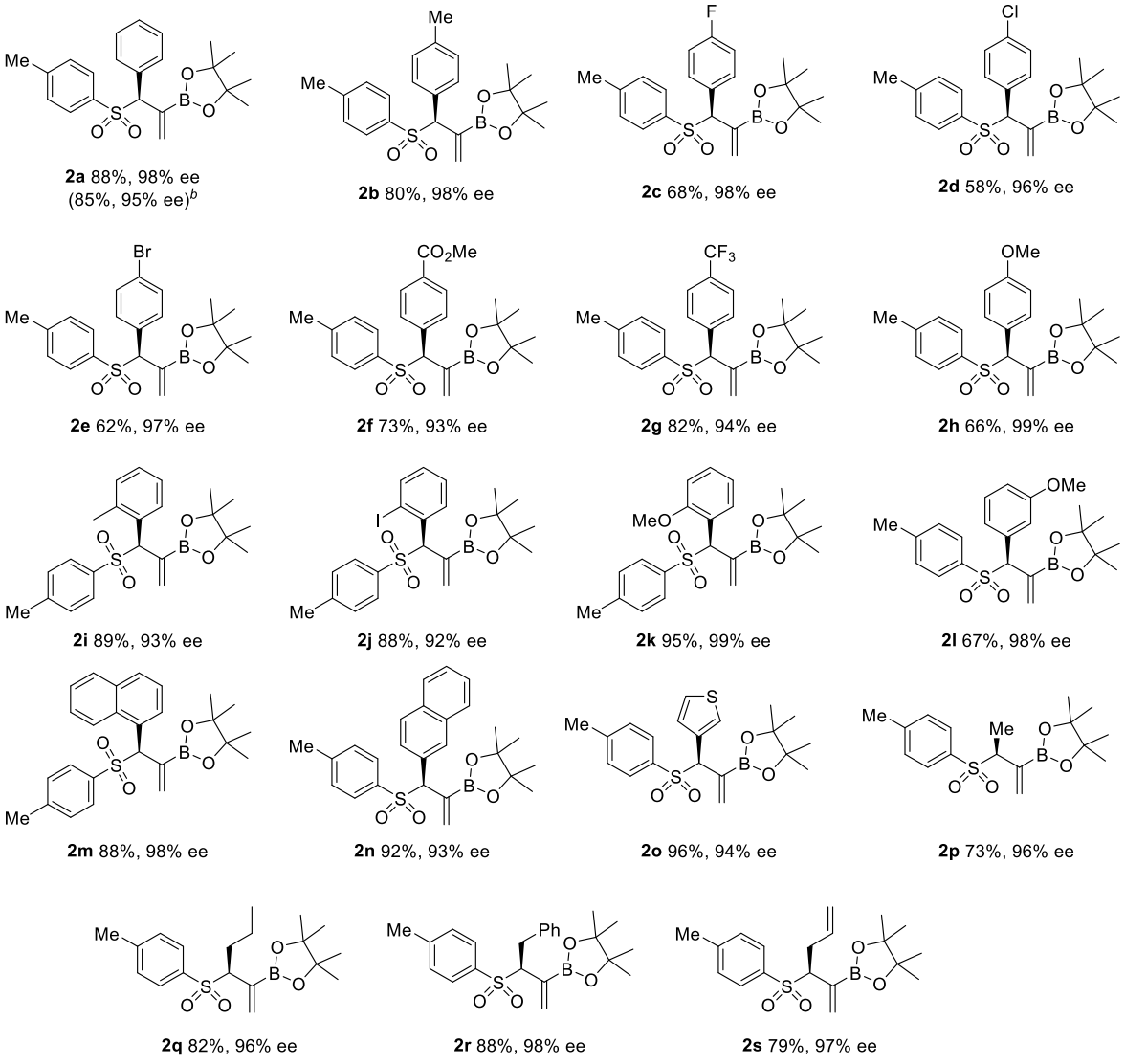
General Considerations. Unless otherwise noted, all the chemicals were purchased from commercial suppliers and were used as received. Solvents were dried and distilled by standard procedures prior to use. All reactions were performed under an inert atmosphere of nitrogen unless otherwise stated. All ^{1}H and ^{13}C{^{1}H} NMR spectra were recorded on JEOL 400 spectrometer. The NMR chemical shift values refer to $CDCl_3$ (δ (^{1}H), 7.26 ppm; δ (^{13}C), 77.16 ppm). Mass spectra were obtained on a micrOTOF-Q II mass spectrometer. Allenyl sulfones were prepared following the literature procedure,[16] and **1a**,[16a] **1p**,[16e] **2a**,[13] **3k−l**,[16] **4l**,[17] **2ac**[18] and **5**[19] are known compounds.

General Procedure A for the Preparation of Allenyl Sulfones. To a solution of sulfonyl chloride (1.1 equiv) and Et_3N (10 equiv) in DCM (0.34 M) was added a solution of triphenylphosphine (1.1 equiv) and alcohol (1 equiv) in DCM (0.34 M) dropwise over a period of 30 min under nitrogen at room temperature. The reaction was monitored by TLC. After the disappearance of starting material, the reaction mixture was quenched with water. The organic layer was separated, and the aqueous phase was extracted with DCM three times, washed with brine, dried over Na_2SO_4 and filtered. Removal of the solvent under vacuum afforded the crude product, which was purified by column chromatography to afford the sulfinate ester.

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Scheme 2. Asymmetric Protoboration of Allenyl Sulfones with Different α-Substituent[a]





[a]Conditions: **1a** (0.2 mmol), B_2pin_2 (0.24 mmol), CuTc (5 mol %), (*S,S*)-Ph-BPE (6 mol %), NaO^{t}Bu (20 mol %), MeOH (0.4 mmol), Toluene (2 mL), 12 h. Isolated yields and ee values were determined by chiral HPLC analysis. [b]1 g scale−up reaction.

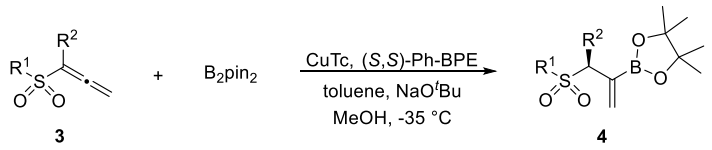
Under nitrogen, to a solution of the sulfinate ester (5 mmol, 1 equiv) in DCM (0.2 M) was added 2 mol % $AgSbF_6$. After 10 min, the reaction mixture was diluted with Et_2O and filtered through a small plug of Celite. Removal of solvent under vacuum afforded the crude product, which was purified by column chromatography to afford the allenyl sulfone.

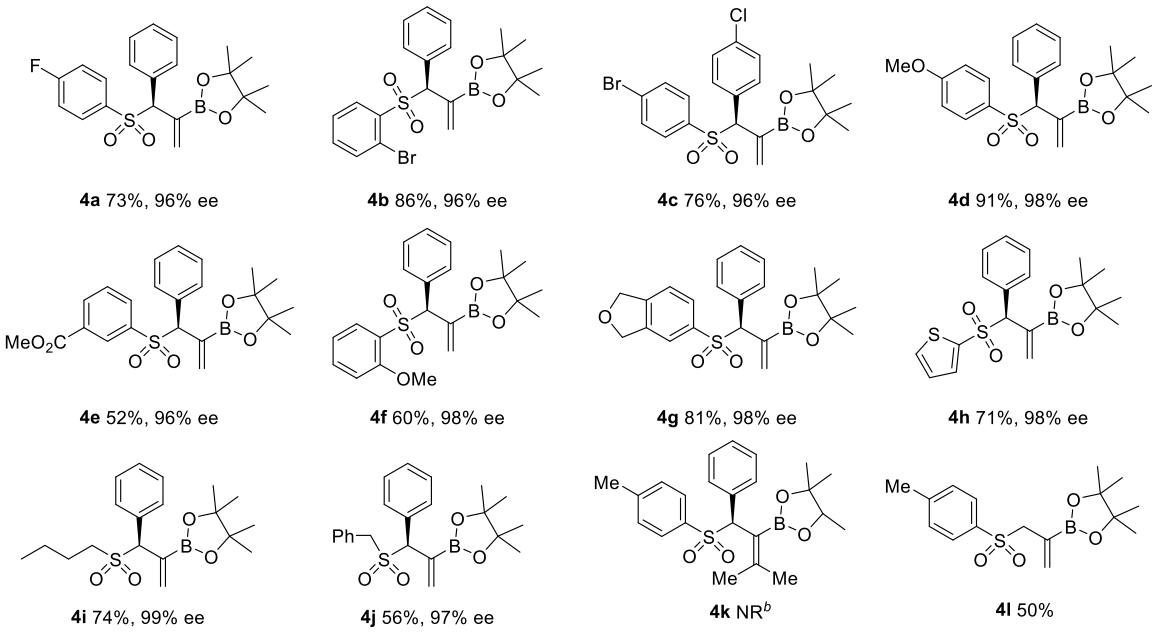
General Procedure B for the Preparation of Allenyl Sulfones. A solution of sulfinyl chloride (2 equiv) in dry diethyl ether (0.4 M) was cooled to −70 °C under nitrogen. A solution of alcohol (1 equiv) and triethylamine (10 equiv) in dry diethyl ether (0.4 M) was added dropwise to the flask at −70 °C. The reaction was stirred for 1 h at −70 °C and allowed to warm to room temperature. After stirring for 2 h at room temperature, the reaction was quenched with water, extract the mixture with ether. Wash the mixture with brine. Removal of the solvent under vacuum afforded the crude product, which was purified by column chromatography to afford the sulfinate ester. Then the sulfinate ester was converted to the allenyl sulfone according to procedure A. Allenyl sulfones **3i** and **3j** were prepared following general procedure B, while the other allenyl sulfones were synthesized using general procedure A.

*Methyl-4-((1-(p-tolyl)propa-1,2-dien-1-yl)sulfonyl)benzene (**1b**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.27 g, 89% yield, pale yellow solid, mp 48−49 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.68 (d, *J* = 8.3 Hz, 2H), 7.32 (d, *J* = 8.2 Hz, 2H), 7.22 (d, *J* = 8.1 Hz, 2H), 7.10 (d, *J* = 8.0 Hz, 2H), 5.51 (s, 2H), 2.37 (s, 3H), 2.30 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 208.9, 144.4, 139.0, 137.3, 129.6, 129.3, 128.8, 128.2, 125.7, 115.2, 83.7, 21.6, 21.3. HRMS (ESI-MS) *m/z* calcd for $C_{17}H_{16}NaO_2S$ $[M + Na]^+$: 307.0763; found, 307.0756.

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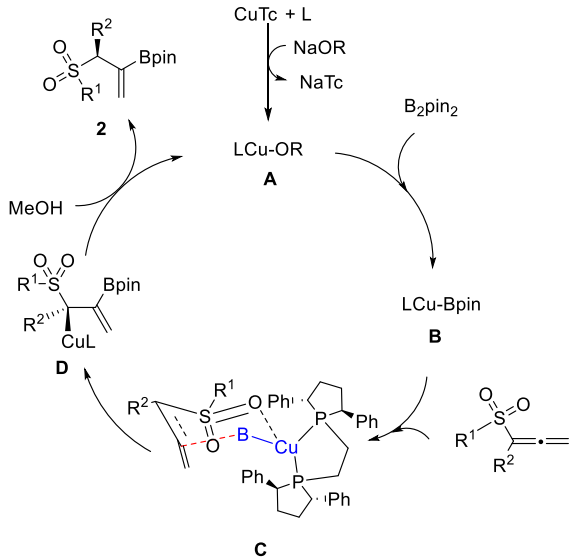
Scheme 3. Asymmetric Protoboration of Allenyl Sulfones[a]





[a]Conditions: **1a** (0.2 mmol), B_2pin_2 (0.24 mmol), CuTc (5 mol %), (*S,S*)-Ph-BPE (6 mol %), NaO^{t}Bu (20 mol %), MeOH (0.4 mmol), toluene (2 mL), 12 h. Isolated yields and ee values were determined by chiral HPLC analysis. [b]−35 °C or room temperature.

Scheme 4. Proposed Mechanism



*1-Fluoro-4-(1-tosylpropa-1,2-dien-1-yl)benzene (**1c**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 5:1) as eluent, 1.18 g, 82% yield, white solid, mp 82−83 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.65 (d, J = 8.3 Hz, 2H), 7.40 (td, J = 5.2, 3.4 Hz, 2H), 7.24 (d, J = 8.0 Hz, 2H), 6.99 (d, J = 8.6, 2.2 Hz, 2H), 5.53 (s, 2H), 2.38 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 208.9, 163.1 (d, J = 250.5 Hz), 144.7, 137.0, 130.9 (d, J = 8.5 Hz), 129.7, 128.3, 124.9 (d, J = 3.4 Hz), 115.8 (d, J = 21.9 Hz), 114.5, 84.1, 21.7. ^{19}F NMR (376 MHz, CDCl3) δ −111.8. HRMS (ESI-MS) m/z calcd for $C_{16}H_{13}FNaO_2S$ [M + Na]$^+$: 311.0512; found, 311.0505.

*1-Chloro-4-(1-tosylpropa-1,2-dien-1-yl)benzene (**1d**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.37 g, 90% yield, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.66 (d, J = 8.3 Hz, 2H), 7.37 (d, J = 8.6 Hz, 2H), 7.27−7.22 (m, 4H), 5.54 (s, 2H), 2.37 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 208.9, 144.8, 137.0, 135.1, 130.2, 129.7, 128.9, 128.3, 127.4, 114.5, 84.4, 21.7. HRMS (ESI-MS) m/z calcd for $C_{16}H_{13}ClNaO_2S$ [M + Na]$^+$: 327.0217 and 329.0191; found, 327.0209 and 329.0182.

*1-Bromo-4-(1-tosylpropa-1,2-dien-1-yl)benzene (**1e**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.50 g, 86% yield, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.67 (d, J = 8.3 Hz, 2H), 7.42 (d, J = 8.7 Hz, 2H), 7.31 (d, J = 8.6 Hz, 2H), 7.24 (d, J = 8.4 Hz, 2H), 5.54 (s, 2H), 2.38 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$) δ 208.9, 144.8, 137.0, 131.9, 130.5, 129.7, 128.3, 127.9, 123.4, 114.6, 84.4, 21.7. HRMS (ESI-MS) m/z calcd for $C_{16}H_{13}BrNaO_2S$ [M + Na]$^+$: 370.9712 and 372.9692; found, 370.9699 and 372.9682.

*Methyl 4-(1-Tosylpropa-1,2-dien-1-yl)benzoate (**1f**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.51 g, 92% yield, pale yellow solid, mp 71−72 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.95 (d, J = 8.5 Hz, 2H), 7.67 (d, J = 8.3 Hz, 2H), 7.53 (d, J = 8.5 Hz, 2H), 7.23 (d, J = 8.2 Hz, 2H), 5.60 (s, 2H), 3.88 (s, 3H), 2.37 (s, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 209.3, 166.6, 144.9, 137.1, 133.5, 130.4, 129.84, 129.75, 128.8, 128.3, 115.0, 84.7, 52.4, 21.7. HRMS (ESI-MS) m/z calcd for $C_{18}H_{16}NaO_4S$ [M + Na]$^+$: 351.0662; found, 351.0645.

*1-Methyl-4-((1-(4-(trifluoromethyl)phenyl)propa-1,2-dien-1-yl)-sulfonyl)benzene (**1g**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.47 g, 87% yield, white solid, mp 64−65 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ

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7.69 (d, J = 8.3 Hz, 2H), 7.60−7.55 (m, 4H), 7.26 (d, J = 8.1 Hz, 2H), 5.60 (s, 2H), 2.40 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.3, 145.0, 137.0, 132.7, 130.9(q, J = 65.4 Hz), 129.8, 129.2, 128.3, 125.6 (q, J = 7.1 Hz), 123.9(q, J = 547.0 Hz), 114.7, 84.7, 21.7. ^{19}F NMR (376 MHz, CDCl3) δ −62.7. HRMS (ESI-MS) m/z calcd for $C_{17}H_{13}F_3NaO_2S$ $[M + Na]^+$: 361.0481; found, 361.0477.

*1-Methoxy-4-(1-tosylpropa-1,2-dien-1-yl)benzene (**1h**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.28 g, 85% yield, colorless oil, ^1H NMR (400 MHz, $CDCl_3$): δ 7.65 (d, J = 8.0 Hz, 2H), 7.33 (d, J = 8.6 Hz, 2H), 7.21 (d, J = 7.9 Hz, 2H), 6.80 (d, J = 8.6 Hz, 2H), 5.49 (s, 2H), 3.74 (s, 3H), 2.35 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.7, 160.0, 144.4, 137.2, 130.3, 129.5, 128.1, 120.6, 114.8, 114.0, 83.7, 55.2, 21.6. HRMS (ESI-MS) m/z calcd for $C_{17}H_{16}NaO_3S$ $[M + Na]^+$: 323.0712; found, 323.0726.

*1-Methyl-2-(1-tosylpropa-1,2-dien-1-yl)benzene (**1i**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.21 g, 85% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.56 (d, J = 8.2 Hz, 2H), 7.24−7.22 (m, 4H), 7.14 (t, J = 8.6 Hz, 2H), 5.42 (s, 2H), 2.41 (s, 3H), 2.01 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.0, 144.6, 138.3, 137.0, 130.8, 130.3, 129.6, 129.5, 128.6, 128.0, 125.8, 112.9, 82.6, 21.8, 19.5. HRMS (ESI-MS) m/z calcd for $C_{17}H_{16}NaO_2S$ $[M + Na]^+$: 307.0763; found, 307.0761.

*1-Iodo-2-(1-tosylpropa-1,2-dien-1-yl)benzene (**1j**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.86 g, 94% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.74 (dd, J = 7.9, 1.0 Hz, 1H), 7.62 (dd, J = 7.8, 1.4 Hz, 1H), 7.54 (d, J = 8.4 Hz, 2H), 7.36 (td, J = 7.8, 1.0 Hz, 1H), 7.21 (d, J = 8.2 Hz, 2H), 7.01 (td, J = 7.7, 1.4 Hz, 1H), 5.55 (s, 2H), 2.39 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.6, 144.8, 139.6, 136.5, 133.0, 131.4, 130.8, 129.6, 128.8, 128.2, 116.3, 101.8, 84.5, 21.8. HRMS (ESI-MS) m/z calcd for $C_{16}H_{13}INaO_2S$ $[M + Na]^+$: 418.9573; found, 418.9585.

*1-Methoxy-2-(1-tosylpropa-1,2-dien-1-yl)benzene (**1k**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 5:1) as eluent, 1.29 g, 86% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.61 (d, J = 8.0 Hz, 2H), 7.34−7.27 (m, 2H), 7.21 (d, J = 7.9 Hz, 2H), 6.92 (t, J = 7.4 Hz, 1H), 6.75 (d, J = 8.3 Hz, 1H), 5.42 (s, 2H), 3.47 (s, 3H), 2.38 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.3, 157.3, 144.1, 137.8, 131.8, 131.0, 129.2, 128.5, 120.5, 117.8, 111.1, 110.5, 82.5, 55.4, 21.7. HRMS (ESI-MS) m/z calcd for $C_{17}H_{16}NaO_3S$ $[M + Na]^+$: 323.0712; found, 323.0709.

*1-Methoxy-3-(1-tosylpropa-1,2-dien-1-yl)benzene (**1l**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 5:1) as eluent, 1.32 g, 88% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.69 (d, J = 8.7 Hz, 2H), 7.23 (d, J = 8.2 Hz, 2H), 7.19 (d, J = 7.7 Hz, 1H), 7.01−6.98 (m, 2H), 6.85−6.82 (m, 1H), 5.54 (s, 2H), 3.74 (s, 3H), 2.37 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.0, 159.5, 144.6, 137.2, 123.0, 129.6, 128.3, 121.3, 115.2, 114.8, 114.2, 84.0, 55.3, 21.7. HRMS (ESI-MS) m/z calcd for $C_{17}H_{16}NaO_3S$ $[M + Na]^+$: 323.0712; found, 323.0711.

*1-(1-Tosylpropa-1,2-dien-1-yl)naphthalene (**1m**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.31 g, 82% yield, pale yellow solid, mp 135−136 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.85 (d, J = 5.3 Hz, 1H), 7.80 (d, J = 7.5 Hz, 1H), 7.74 (d, J = 7.8 Hz, 1H), 7.57 (d, J = 7.3 Hz, 2H), 7.46−7.37 (m, 4H), 7.14 (d, J = 7.2 Hz, 2H), 5.50 (s, 2H), 2.34 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.8, 144.6, 136.6, 133.6, 132.0, 130.0, 129.5, 129.1, 128.6, 128.3, 126.7, 126.1, 126.0, 125.0, 124.9, 112.2, 82.8, 21.7. HRMS (ESI-MS) m/z calcd for $C_{20}H_{16}NaO_2S$ $[M + Na]^+$: 343.0763; found, 343.0761.

*2-(1-Tosylpropa-1,2-dien-1-yl)naphthalene (**1n**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.38 g, 86% yield, pale yellow solid, mp 96−97 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 8.05 (s, 1H), 7.82−7.76 (m, 3H), 7.74 (d, J = 8.3 Hz, 2H), 7.51−7.46 (m, 3H), 7.21 (d, J = 8.2 Hz, 2H), 5.62 (s, 2H), 2.35 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.5, 144.6, 137.3, 133.14, 133.08, 129.7, 128.46, 128.42, 128.38, 128.28, 127.7, 126.9, 126.6, 126.2, 126.0, 115.6, 84.3, 21.7. HRMS (ESI-MS) m/z calcd for $C_{20}H_{16}NaO_2S$ $[M + Na]^+$: 343.0763; found, 343.0759.

*3-(1-Tosylpropa-1,2-dien-1-yl)thiophene (**1o**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.15 g, 83% yield, pale yellow solid, mp 81−82 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.75 (d, J = 8.3 Hz, 2H), 7.60 (d, J = 1.8 Hz, 1H), 7.27−7.24 (m, 3H), 7.06 (dd, J = 5.0, 1.4 Hz, 1H), 5.58 (s, 2H), 2.39 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.0, 144.6, 137.4, 129.7, 128.1, 127.74, 127.68, 125.9, 124.8, 111.8, 84.6, 21.7. HRMS (ESI-MS) m/z calcd for $C_{14}H_{12}NaO_2S_2$ $[M + Na]^+$: 299.0171; found, 299.0170.

*1-(Hexa-1,2-dien-3-ylsulfonyl)-4-methylbenzene (**1q**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.00 g, 85% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.76 (d, J = 8.2 Hz, 2H), 7.32 (d, J = 8.1 Hz, 2H), 5.34 (t, J = 3.5 Hz, 2H), 2.43 (s, 2H), 2.21−2.16 (m, 2H), 1.44 (q, J = 7.4 Hz, 2H), 0.87 (t, J = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 207.7, 144.5, 137.4, 129.8, 128.2, 113.4, 84.3, 28.7, 21.8, 20.9, 13.5. HRMS (ESI-MS) m/z calcd for $C_{13}H_{16}NaO_2S$ $[M + Na]^+$: 259.0763; found, 259.0761.

*1-Methyl-4-((1-phenylbuta-2,3-dien-2-yl)sulfonyl)benzene (**1r**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.25 g, 88% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.72 (d, J = 8.2 Hz, 2H), 7.28 (d, J = 8.1 Hz, 2H), 7.23−7.15 (m, 3H), 7.07−7.04 (m, 2H), 5.24 (t, J = 3.1 Hz, 2H), 3.59 (t, J = 3.2 Hz, 2H), 2.43 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.5, 144.5, 137.3, 136.4, 129.7, 129.0, 128.4, 128.2, 126.9, 113.5, 84.6, 33.6, 21.7. HRMS (ESI-MS) m/z calcd for $C_{17}H_{16}NaO_2S$ $[M + Na]^+$: 307.0763; found, 307.0760.

*1-(Hexa-1,2,5-trien-3-ylsulfonyl)-4-methylbenzene (**1s**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 0.99 g, 85% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.76 (d, J = 8.3 Hz, 2H), 7.32 (d, J = 8.2 Hz, 2H), 5.69−5.59 (m, 1H), 5.35 (t, J = 3.2 Hz, 2H), 5.07−5.02 (m, 2H), 3.00−2.97 (m, 2H), 2.43 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.0, 144.7, 137.2, 132.6, 129.8, 128.3, 118.0, 112.1, 84.6, 31.4, 21.7. HRMS (ESI-MS) m/z calcd for $C_{13}H_{14}NaO_2S$ $[M + Na]^+$: 257.0607; found, 257.0608.

*1-Fluoro-4-((1-phenylpropa-1,2-dien-1-yl)sulfonyl)benzene (**3a**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.26 g, 92% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.82−7.79 (m, 2H), 7.42 (m, 2H), 7.33−7.29 (m, 3H), 7.13−7.08 (m, 2H), 5.56 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.0, 165.7 (d, J = 257.3 Hz), 136.1 (d, J = 3.0 Hz), 131.2 (d, J = 9.6 Hz), 129.2, 129.0, 128.8, 128.5, 116.3 (d, J = 22.7 Hz), 115.2, 84.2. ^{19}F NMR (376 MHz, CDCl3) δ −103.4. HRMS (ESI-MS) m/z calcd for $C_{15}H_{11}FKO_2S$ $[M + K]^+$: 313.0095; found, 313.0091.

*1-Bromo-2-((1-phenylpropa-1,2-dien-1-yl)sulfonyl)benzene (**3b**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.41 g, 84% yield, pale yellow solid, mp 83−84 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 8.01 (dd, J = 7.0, 2.0 Hz, 1H), 7.69 (dd, J = 7.0, 2.2 Hz, 1H), 7.43−7.34 (m, 4H), 7.27−7.26 (m, 3H), 5.65 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 210.5, 138.2, 135.4, 134.6, 132.6, 129.1, 128.8, 128.7, 127.7, 121.2, 112.9, 84.9. HRMS (ESI-MS) m/z calcd for $C_{15}H_{11}BrNaO_2S$ $[M + Na]^+$: 356.9556 and 358.9535; found, 356.9557 and 358.9538.

*1-Bromo-4-((1-(4-chlorophenyl)propa-1,2-dien-1-yl)sulfonyl)-benzene (**3c**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.72 g, 93% yield, white solid, mp 87−88 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.64 (d, J = 8.8 Hz, 2H), 7.60 (d, J = 8.7 Hz, 2H), 7.37 (d, J = 8.6 Hz, 2H), 7.30 (d, J = 8.7 Hz, 2H), 5.60 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.2, 139.0, 135.5, 132.5, 130.3, 129.8, 129.1, 127.0, 114.3, 84.7. HRMS (ESI-MS) m/z calcd for $C_{15}H_{10}BrClNaO_2S$ $[M + Na]^+$: 390.9166, 392.9143 and 394.9112; found, 390.9156, 392.9130 and 394.9109.

*1-Methoxy-4-((1-phenylpropa-1,2-dien-1-yl)sulfonyl)benzene (**3d**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 9:1) as eluent, 1.25 g, 87% yield, white solid, mp

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91−92 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.71 (d, J = 8.8 Hz, 2H), 7.44−7.41 (m, 2H), 7.30−7.28 (m, 3H), 6.88 (d, J = 8.9 Hz, 2H), 5.50 (s, 2H), 3.80 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.6, 163.6, 131.6, 130.5, 128.9, 128.8, 128.6, 115.5, 114.1, 83.8, 55.6. HRMS (ESI-MS) m/z calcd for $C_{16}H_{14}NaO_3S$ $[M + Na]^+$: 309.0556; found, 309.0562.

*Methyl 3-((1-phenylpropa-1,2-dien-1-yl)sulfonyl)benzoate (**3e**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 6:1) as eluent, 1.56 g, 99% yield, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.45 (s, 1H), 8.18 (d, J = 7.8 Hz, 1H), 7.93 (d, J = 8.0 Hz, 1H), 7.50 (t, J = 7.8 Hz, 1H), 7.41−7.38 (m, 2H), 7.28−7.26 (m, 3H), 5.56 (s, 2H), 3.88 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.1, 165.2, 140.6, 134.3, 132.2, 131.1, 129.2, 129.13, 129.10, 128.8, 128.7, 128.2, 114.8, 84.4, 52.5. HRMS (ESI-MS) m/z calcd for $C_{17}H_{14}NaO_4S$ $[M + Na]^+$: 337.0505; found, 337.0510.

*1-Methoxy-2-((1-phenylpropa-1,2-dien-1-yl)sulfonyl)benzene (**3f**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 1.23 g, 86% yield, white solid, mp 78−79 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.92 (d, J = 7.6 Hz, 1H), 7.50 (t, J = 7.5 Hz, 1H), 7.42−7.40 (m, 2H), 7.28−7.22 (m, 3H), 7.00 (t, J = 7.6 Hz, 1H), 6.94 (d, J = 8.4 Hz, 1H), 5.57 (s, 2H), 3.86 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 210.3, 157.4, 135.6, 131.0, 129.0, 128.7, 128.6, 128.5, 127.5, 120.4, 114.6, 112.2, 83.8, 56.2. HRMS (ESI-MS) m/z calcd for $C_{16}H_{14}NaO_3S$ $[M + Na]^+$: 309.0556; found, 309.0559.

*5-((1-Phenylpropa-1,2-dien-1-yl)sulfonyl)-1,3-dihydroisobenzofuran (**3g**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 1.25 g, 84% yield, white solid, mp 94−95 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.56 (s, 1H), 7.53 (d, J = 8.5 Hz, 1H), 7.41−7.39 (m, 2H), 7.27−7.25 (m, 3H), 6.71 (d, J = 8.4 Hz, 1H), 5.48 (s, 2H), 4.59 (t, J = 8.8 Hz, 2H), 3.14 (t, J = 8.7 Hz, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.6, 164.6, 131.6, 130.1, 128.9, 128.6, 128.3, 125.4, 115.6, 109.3, 83.8, 72.4, 28.9. HRMS (ESI-MS) m/z calcd for $C_{17}H_{14}NaO_3S$ $[M + Na]^+$: 321.0556; found, 321.0550.

*2-((1-Phenylpropa-1,2-dien-1-yl)sulfonyl)thiophene (**3h**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.30 g, 99% yield, white solid, mp 50−51 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.62 (dd, J = 4.9, 1.2 Hz, 1H), 7.54 (dd, J = 3.9, 1.4 Hz, 1H), 7.49−7.46 (m, 2H), 7.36−7.33 (m, 3H), 7.03 (dd, J = 4.8, 3.8 Hz, 1H), 5.59 (s, 2H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 208.7, 141.1, 134.6, 134.2, 129.3, 129.1, 128.8, 128.6, 127.6, 116.1, 84.4. HRMS (ESI-MS) m/z calcd for $C_{13}H_{10}NaO_2S_2$ $[M + Na]^+$: 285.0014; found, 285.0012.

*(1-(Butylsulfonyl)propa-1,2-dien-1-yl)benzene (**3i**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.02 g, 86% yield, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.62−7.60 (m, 2H), 7.43−7.36 (m, 3H), 5.64 (s, 2H), 3.02 (t, J = 8.0 Hz, 2H), 1.76 (p, J = 7.9 Hz, 2H), 1.42−1.33 (m, 2H), 0.87 (t, J = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 209.8, 129.2, 129.0, 128.8, 128.6, 112.7, 84.1, 53.5, 24.4, 21.5, 13.6. HRMS (ESI-MS) m/z calcd for $C_{13}H_{16}NaO_2S$ $[M + Na]^+$: 259.0763; found, 259.0755.

*(1-(Benzylsulfonyl)propa-1,2-dien-1-yl)benzene (**3j**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 10:1) as eluent, 1.12 g, 83% yield, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.59−7.56 (m, 2H), 7.44−7.38 (m, 3H), 7.37−7.31 (m, 3H), 7.26−7.24 (m, 2H), 5.32 (s, 2H), 4.26 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 211.1, 131.1, 129.2, 129.07, 129.02, 128.8, 128.7, 128.2, 111.3, 83.5, 60.5. HRMS (ESI-MS) m/z calcd for $C_{16}H_{14}NaO_2S$ $[M + Na]^+$: 293.0607; found, 293.0593.

General Procedure for Copper-Catalyzed Asymmetric Protoboration of Allenyl Sulfones. To a 25 mL Schlenk tube were added CuTc (1.9 mg, 5 mol %), (*S*,*S*)-Ph-BPE (6.1 mg, 6 mol %), NaO^{t}Bu (3.8 mg, 20 mol %) and toluene (2 mL) under nitrogen. The mixture was stirred at rt for 30 min and cool to −35 °C. Then B_2pin_2 (0.24 mmol) was added. After 5 min, allenyl sulfone (0.2 mmol) and MeOH (16 μL, 2.0 equiv) were added sequentially. The reaction was stirred at −35 °C for 12 h. The reaction mixture was quenched with NH_4Cl solution at −35 °C and extracted with ethyl acetate (10 mL × 3). The combined organic phase was washed with brine and dried over anhydrous Na_2SO_4, filtered, and concentrated. The resultant residue was purified by column chromatography to give the product.

Procedure for One Gram Scale−Up Reaction. To a 100 mL Schlenk flask were added CuTc (35.3 mg, 5 mol %), (*S*,*S*)-Ph-BPE (112.5 mg, 6 mol %), NaO^{t}Bu (71.1 mg, 20 mol %) and toluene (37 mL) under nitrogen. The mixture was stirred at rt for 30 min and cool to −35 °C. Then B_2pin_2 (1.13 g, 4.4 mmol) was added. After 5 min, allenyl sulfone **1a** (1.0 g, 3.70 mmol) and MeOH (300 μL, 2.0 equiv) were added sequentially. The reaction was stirred at −35 °C for 12 h. The reaction mixture was quenched with NH_4Cl solution and extracted with ethyl acetate (30 mL × 3). The combined organic phase was washed with brine and dried over anhydrous Na_2SO_4, filtered, and concentrated. The resultant residue was purified by column chromatography to give the product **2a** (1.25 g, 85% yield, 95% ee).

*(R)-4,4,5,5-Tetramethyl-2-(3-phenyl-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2a**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 70.1 mg, 88% yield, 98% ee, white solid, mp 103−104 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.53 (d, J = 8.2 Hz, 2H), 7.36−7.33 (m, 2H), 7.27−7.22 (m, 3H), 7.18 (d, J = 8.5 Hz, 2H), 6.55 (s, 1H), 6.26 (d, J = 1.7 Hz, 1H), 5.12 (s, 1H), 2.36 (s, 3H), 2.31 (s, 3H), 1.14 (s, 6H), 1.10 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 144.2, 136.2, 135.0, 132.6, 129.29, 129.26, 128.5, 128.3, 84.2, 72.1, 24.8, 24.5, 21.7. $[\alpha]_D^{20}$ = −64.1 (c = 1.6, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{22}H_{27}BNaO_4S$ $[M + Na]^+$: 421.1615; found, 421.1608. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.0 min (minor), 10.6 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-(p-tolyl)-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2b**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 65.9 mg, 80% yield, 98% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.55 (d, J = 8.3 Hz, 2H), 7.25 (d, J = 8.0 Hz, 2H), 7.19 (d, J = 8.1 Hz, 2H), 7.07 (d, J = 8.0 Hz, 2H), 6.52 (s, 1H), 6.24 (d, J = 2.0 Hz, 1H), 5.11 (s, 1H), 2.37 (s, 3H), 2.31 (s, 3H), 1.14 (s, 6H), 1.11 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 144.1, 138.3, 136.3, 135.1, 130.5, 129.4, 129.3, 129.1, 84.2, 71.7, 24.8, 24.6, 21.7, 21.2. $[\alpha]_D^{20}$ = +239.7 (c = 1.7, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BKO_4S$ $[M + K]^+$: 451.1515; found, 451.1498. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 5.3 min (minor), 7.3 min (major).

*(R)-2-(3-(4-Fluorophenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2c**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 2:1) as eluent, 56.6 mg, 68% yield, 98% ee, pale yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.53 (d, J = 8.3 Hz, 2H), 7.33−7.30 (m, 2H), 7.20 (d, J = 7.9 Hz, 2H), 6.97−6.92 (m, 2H), 6.53 (s, 1H), 6.26 (d, J = 1.8 Hz, 1H), 5.10 (s, 1H), 2.38 (s, 3H), 1.14 (s, 6H), 1.10 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 162.9 (d, J = 249.0 Hz), 144.4, 136.0, 134.9, 132.4 (d, J = 8.4 Hz), 129.4, 129.2, 128.4(d, J = 3.1 Hz), 115.3(d, J = 21.6 Hz), 84.3, 71.3, 24.8, 24.5, 21.7. ^{19}F NMR (376 MHz, CDCl3) δ −113.2. HRMS (ESI-MS) m/z calcd for $C_{22}H_{27}BFO_4S$ $[M + H]^+$: 417.1706; found, 417.1706. $[\alpha]_D^{20}$ = +293.7 (c = 1.3, $CHCl_3$). HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 5.1 min (minor), 8.5 min (major).

*(R)-2-(3-(4-Chlorophenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2d**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 2:1) as eluent, 50.2 mg, 58% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.54 (d, J = 8.3 Hz, 2H), 7.29 (d, J = 8.6 Hz, 2H), 7.25−7.20 (m, 4H), 6.52 (s, 1H), 6.26 (d, J = 1.8 Hz, 1H), 5.09 (s, 1H), 2.38 (s, 3H), 1.15 (s, 6H), 1.10 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 144.5, 135.9, 135.2, 134.6, 131.9, 131.2, 129.4, 129.2, 128.5, 84.3, 71.3, 24.8, 24.5, 21.7. $[\alpha]_D^{20}$ = +219.8 (c = 1.2, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{22}H_{26}BClNaO_4S$ $[M + Na]^+$: 455.1229 and 457.1207; found, 455.1223 and 457.1199. HPLC (AD-

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H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.1 min (minor), 10.0 min (major).

*(R)-2-(3-(4-Bromophenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2e**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 59.2 mg, 62% yield, 97% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.54 (d, J = 8.2 Hz, 2H), 7.39 (d, J = 8.5 Hz, 2H), 7.24−7.20 (m, 4H), 6.51 (s, 1H), 6.26 (d, J = 1.7 Hz, 1H), 5.08 (s, 1H), 2.39 (s, 3H), 1.15 (s, 6H), 1.10 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.5, 135.9, 135.3, 132.2, 131.8, 131.5, 129.5, 129.2, 122.9, 84.3, 71.4, 24.8, 24.5, 21.7. $[\alpha]_D^{20}$ = +321.3 (c = 3.1, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{22}H_{26}BBrNaO_4S$ $[M + Na]^+$: 499.0724 and 501.0706; found, 499.0716 and 501.0704. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 5.7 min (minor), 6.8 min (major).

*(R)-Methyl 4-(2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-tosylallyl)benzoate (**2f**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 66.6 mg, 73% yield, 93% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.92 (d, J = 8.4 Hz, 2H), 7.52 (d, J = 8.3 Hz, 2H), 7.42 (d, J = 8.4 Hz, 2H), 7.19 (d, J = 8.1 Hz, 2H), 6.56 (s, 1H), 6.29 (d, J = 1.7 Hz, 1H), 5.17 (s, 1H), 3.90 (s, 3H), 2.38 (s, 3H), 1.14 (s, 6H), 1.09 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 166.8, 144.5, 137.9, 135.9, 135.5, 130.6, 130.1, 129.5, 129.4, 129.2, 84.3, 71.9, 52.3, 24.8, 24.5, 21.7. $[\alpha]_D^{20}$ = +322.7 (c = 1.1, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{24}H_{29}BNaO_6S$ $[M + Na]^+$: 479.1675; found, 479.1676. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.7 min (minor), 13.4 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-tosyl-3-(4-(trifluoromethyl)phenyl)-prop-1-en-2-yl)-1,3,2-dioxaborolane (**2g**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 76.5 mg, 82% yield, 94% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (d, J = 8.2 Hz, 2H), 7.52−7.50 (m, 4H), 7.22 (d, J = 8.1 Hz, 2H), 6.54 (s, 1H), 6.29 (d, J = 1.5 Hz, 1H), 5.19 (s, 1H), 2.39 (s, 3H), 1.13 (s, 6H), 1.10 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.7, 136.9, 135.84, 135.81, 130.9, 129.5, 129.2, 125.3 (q, J = 3.6 Hz), 84.4, 71.5, 24.8, 24.5, 21.7. ^{19}F NMR (376 MHz, $CDCl_3$): δ −62.5. $[\alpha]_D^{20}$ = +91.1 (c = 2.6, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{26}BF_3NaO_4S$ $[M + Na]^+$: 489.1493; found, 489.1493. HPLC (IC−H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.7 min (minor), 10.1 min (major).

*(R)-2-(3-(4-Methoxyphenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2h**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 2:1) as eluent, 56.5 mg, 66% yield, 99% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.53 (d, J = 8.1 Hz, 2H), 7.26 (d, J = 8.6 Hz, 2H), 7.20 (d, J = 8.0 Hz, 2H), 6.79 (d, J = 8.6 Hz, 2H), 6.53 (s, 1H), 6.24 (d, J = 1.2 Hz, 1H), 5.08 (s, 1H), 3.78 (s, 3H), 2.38 (s, 3H), 1.15 (s, 6H), 1.11 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 159.8, 144.1, 136.2, 134.7, 131.9, 129.29, 129.25, 124.3, 113.7, 84.2, 71.4, 55.3, 24.8, 24.6, 21.7. $[\alpha]_D^{20}$ = +261.1 (c = 2.2, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BNaO_5S$ $[M + Na]^+$: 451.1725; found, 451.1723. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 8.5 min (minor), 14.8 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-(o-tolyl)-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2i**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 73.4 mg, 89% yield, 93% ee, pale yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.79 (d, J = 8.8 Hz, 1H), 7.50 (d, J = 8.3 Hz, 2H), 7.23−7.13 (m, 4H), 6.99 (d, J = 7.4 Hz, 1H), 6.64 (s, 1H), 6.24 (d, J = 1.8 Hz, 1H), 5.42 (s, 1H), 2.37 (s, 3H), 1.98 (s, 3H), 1.11 (s, 6H), 1.05 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.2, 138.0, 136.7, 134.5, 131.5, 130.2, 130.1, 129.3, 129.1, 128.3, 126.1, 84.1, 66.9, 24.9, 24.3, 21.7, 19.3. $[\alpha]_D^{20}$ = +266.4 (c = 1.5, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BNaO_4S$ $[M + Na]^+$: 435.1776; found, 435.1775. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.3 min (minor), 9.8 min (major).

*(R)-2-(3-(2-Iodophenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2j**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 92.3 mg, 88% yield, 92% ee, white solid, mp 108−109 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.95 (dd, J = 7.9, 1.3 Hz, 1H), 7.71 (dd, J = 7.9, 0.8 Hz, 1H), 7.54 (d, J = 8.2 Hz, 2H), 7.38−7.35 (m, 1H), 7.20 (d, J = 8.1 Hz, 2H), 6.94 (td, J = 7.6, 1.3 Hz, 1H), 6.51 (s, 1H), 6.26 (d, J = 1.4 Hz, 1H), 5.66 (s, 1H), 2.39 (s, 3H), 1.14 (s, 6H), 1.08 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.5, 139.7, 136.3, 136.1, 135.3, 131.3, 129.9, 129.4, 128.2, 103.9, 84.2, 74.9, 25.0, 24.3, 21.7. $[\alpha]_D^{20}$ = +105.5 (c = 2.2, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{22}H_{27}BIO_4S$ $[M + H]^+$: 525.0766; found, 525.0767. HPLC (OD-H, *n*-hexane/isopropanol = 95/5, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.6 min (minor), 9.3 min (major).

*(R)-2-(3-(2-Methoxyphenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2k**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 81.4 mg, 95% yield, >99% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76 (dd, J = 7.7, 1.6 Hz, 1H), 7.52 (d, J = 8.2 Hz, 2H), 7.21 (td, J = 8.3, 1.7 Hz, 1H), 7.15 (d, J = 8.0 Hz, 2H), 6.94 (td, J = 7.6, 1.0 Hz, 1H), 6.64 (d, J = 8.2 Hz, 1H), 6.48 (s, 1H), 6.21 (d, J = 1.9 Hz, 1H), 5.82 (s, 1H), 3.45 (s, 3H), 2.35 (s, 3H), 1.13 (s, 6H), 1.06 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 157.2, 143.8, 136.8, 134.6, 130.8, 129.5, 129.3, 128.9, 121.8, 120.4, 110.4, 83.9, 63.4, 55.3, 24.9, 24.3, 21.6. $[\alpha]_D^{20}$ = +25.1 (c = 2.5, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BNaO_5S$ $[M + Na]^+$: 451.1725; found, 451.1726. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 14.8 min (major).

*(R)-2-(3-(3-Methoxyphenyl)-3-tosylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**2l**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 57.4 mg, 67% yield, 98% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (d, J = 8.2 Hz, 2H), 7.20 (d, J = 8.1 Hz, 2H), 7.15 (d, J = 8.2 Hz, 1H), 6.94−6.93 (m, 2H), 6.83−6.81 (m, 1H), 6.53 (s, 1H), 6.26 (d, J = 1.4 Hz, 1H), 5.10 (s, 1H), 3.74 (s, 3H), 2.38 (s, 3H), 1.15 (s, 6H), 1.12 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 159.4, 144.2, 136.2, 135.3, 134.0, 129.3, 123.0, 115.8, 114.4, 84.2, 72.0, 55.3, 24.8, 24.5, 21.7. $[\alpha]_D^{20}$ = +216.5 (c = 1.0, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BNaO_5S$ $[M + Na]^+$: 451.1725; found, 451.1716. HPLC (AD-H, *n*-hexane/isopropanol = 50/50, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.7 min (minor), 17.1 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-(naphthalen-1-yl)-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2m**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 78.9 mg, 88% yield, 98% ee, pale yellow oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.29 (d, J = 7.3 Hz, 1H), 7.90 (d, J = 8.1 Hz, 1H), 7.80−7.78 (m, 2H), 7.66 (d, J = 8.3 Hz, 2H), 7.52 (t, J = 7.9 Hz, 1H), 7.42−7.35 (m, 2H), 7.16 (d, J = 8.1 Hz, 2H), 6.52 (s, 1H), 6.30 (d, J = 1.8 Hz, 1H), 6.07 (s, 1H), 2.32 (s, 3H), 1.10 (s, 6H), 1.06 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.2, 137.1, 136.8, 133.9, 131.8, 130.2, 129.4, 128.9, 128.8, 127.2, 126.2, 125.5, 125.3, 122.9, 84.2, 65.9, 25.0, 24.3, 21.6. $[\alpha]_D^{20}$ = −238.3 (c = 1.3, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{26}H_{29}BNaO_4S$ $[M + Na]^+$: 471.1777; found, 471.1781. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.2 min (minor), 16.4 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-(naphthalen-2-yl)-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2n**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 2:1) as eluent, 82.5 mg, 92% yield, 93% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.84 (s, 1H), 7.81−7.74 (m, 3H), 7.57 (d, J = 8.2 Hz, 2H), 7.51 (dd, J = 8.6, 1.6 Hz, 1H), 7.46 (t, J = 3.8 Hz, 2H), 7.16 (d, J = 8.1 Hz, 2H), 6.65 (s, 1H), 6.33 (d, J = 1.7 Hz, 1H), 5.34 (s, 1H), 2.35 (s, 3H), 1.16 (s, 6H), 1.11 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.2, 136.1, 135.3, 133.10, 133.05, 130.2, 130.1, 129.3, 129.2, 128.3, 127.9, 127.8, 127.6, 126.5, 126.2, 84.2, 72.1, 24.8, 24.5, 21.6. $[\alpha]_D^{20}$ = +425.7 (c = 1.9, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{26}H_{29}BNaO_4S$ $[M + Na]^+$: 471.1777; found, 471.1778. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 9.6 min (minor), 17.0 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-(thiophen-3-yl)-3-tosylprop-1-en-2-yl)-1,3,2-dioxaborolane (**2o**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 77.6 mg,

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96% yield, 94% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (d, J = 8.2 Hz, 2H), 7.29−7.28 (m, 1H), 7.27−7.25 (m, 1H), 7.22 (d, J = 8.0 Hz, 2H), 7.16−7.15 (m, 1H), 6.39 (s, 1H), 6.29 (d, J = 1.7 Hz, 1H), 5.32 (s, 1H), 2.39 (s, 3H), 1.19 (s, 6H), 1.17 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.3, 136.1, 135.8, 132.5, 129.34, 129.30, 129.1, 126.3, 125.4, 84.2, 67.8, 24.72, 24.68, 21.7. $[\alpha]_D^{20}$ = +13.4 (c = 1.3, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{20}H_{25}BNaO_4S_2$ [M + Na]$^+$: 427.1183; found, 427.1179. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.2 min (minor), 11.2 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-tosylbut-1-en-2-yl)-1,3,2-dioxaborolane (**2p**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 49.1 mg, 73% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.67 (d, J = 8.1 Hz, 2H), 7.28 (d, J = 8.0 Hz, 2H), 6.07 (d, J = 1.6 Hz, 1H), 5.82 (s, 1H), 4.07 (q, J = 7.1 Hz, 1H), 2.40 (s, 3H), 1.47 (d, J = 7.2 Hz, 3H), 1.18 (s, 12H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.3, 134.8, 134.6, 129.8, 129.3, 84.0, 62.6, 24.8, 24.7, 21.7, 13.2. $[\alpha]_D^{20}$ = −401.0 (c = 2.0, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{17}H_{25}BNaO_4S$ [M + Na]$^+$: 359.1462; found, 359.1462. HPLC (AD-H, n-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 8.3 min (minor), 10.1 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-tosylhex-1-en-2-yl)-1,3,2-dioxaborolane (**2q**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 59.7 mg, 82% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.64 (d, J = 8.1 Hz, 2H), 7.25 (d, J = 8.1 Hz, 2H), 6.11 (d, J = 2.2 Hz, 1H), 5.87 (d, J = 1.8 Hz, 1H), 3.89 (dd, J = 11.6, 3.7 Hz, 1H), 2.38 (s, 3H), 2.13−2.04 (m, 1H), 1.98−1.88 (m, 1H), 1.41−1.30 (m, 1H), 1.26−1.19 (m, 1H), 1.11 (s, 6H), 1.10 (s, 6H), 0.85 (t, J = 7.4 Hz, 3H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.1, 136.3, 135.4, 129.8, 129.3, 83.9, 67.6, 28.8, 24.6, 21.7, 20.0, 13.7. $[\alpha]_D^{20}$ = −220.8 (c = 1.0, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{19}H_{29}BNaO_4S$ [M + Na]$^+$: 387.1775; found, 387.1770. HPLC (AD-H, n-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.3 min (minor), 7.8 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(4-phenyl-3-tosylbut-1-en-2-yl)-1,3,2-dioxaborolane (**2r**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 72.6 mg, 88% yield, 98% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.76 (d, J = 8.2 Hz, 2H), 7.32 (d, J = 8.0 Hz, 2H), 7.26−7.22 (m, 2H), 7.18−7.16 (m, 3H), 6.09 (d, J = 2.0 Hz, 1H), 5.97 (s, 1H), 4.26 (dd, J = 12.0, 3.2 Hz, 1H), 3.66 (dd, J = 13.8, 3.3 Hz, 1H), 3.32 (t, J = 12.4 Hz, 1H), 2.44 (s, 3H), 1.12 (s, 6H), 1.08 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.3, 137.4, 137.0, 135.3, 129.6, 129.4, 129.3, 128.3, 126.4, 83.8, 68.9, 32.9, 24.6, 24.4, 21.6. $[\alpha]_D^{20}$ = +137.7 (c = 3.3, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{29}BNaO_4S$ [M + Na]$^+$: 435.1776; found, 435.1769. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 4.8 min (minor), 6.1 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-tosylhexa-1,5-dien-2-yl)-1,3,2-dioxaborolane (**2s**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 57.2 mg, 79% yield, 97% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.66 (d, J = 8.3 Hz, 2H), 7.27 (d, J = 8.2 Hz, 2H), 6.11 (d, J = 2.1 Hz, 1H), 5.83 (d, J = 1.5 Hz, 1H), 5.67−5.57 (m, 1H), 5.07−4.99 (m, 2H), 3.95 (dd, J = 11.4, 3.8 Hz, 1H), 2.94−2.87 (m, 1H), 2.76−2.68 (m, 1H), 2.39 (s, 3H), 1.14 (s, 6H), 1.13 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 144.3, 136.6, 135.1, 133.8, 129.8, 129.4, 117.8, 83.9, 67.7, 31.4, 24.7, 24.6, 21.7. $[\alpha]_D^{20}$ = −106.8 (c = 1.0, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{19}H_{27}BNaO_4S$ [M + Na]$^+$: 385.1619; found, 385.1612. HPLC (AD-H, n-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.0 min (minor), 9.0 min (major).

*(R)-2-(3-((4-Fluorophenyl)sulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4a**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 58.7 mg, 73% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.66−7.61 (m, 2H), 7.33−7.30 (m, 2H), 7.27−7.22 (m, 3H), 7.07−7.03 (m, 2H), 6.56 (s, 1H), 6.27 (d, J = 1.8 Hz, 1H), 5.10 (s, 1H), 1.14 (s, 6H), 1.10 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 165.7 (d, J = 256.9 Hz), 135.3, 135.1 (d, J = 2.8 Hz), 132.3, 132.1 (d, J = 9.6 Hz), 130.6, 128.7, 128.5, 115.9 (d, J = 22.6 Hz), 84.3, 72.3, 24.8, 24.6. ^{19}F NMR (376 MHz, CDCl3): δ −104.1. $[\alpha]_D^{20}$ = +219.9 (c = 1.7, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{21}H_{24}BFNaO_4S$ [M + Na]$^+$: 425.1368; found, 425.1362. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 4.8 min (minor), 5.9 min (major).

*(R)-2-(3-((2-Bromophenyl)sulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4b**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 79.7 mg, 86% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.81 (dd, J = 7.6, 1.8 Hz, 1H), 7.71−7.69 (m, 1H), 7.46−7.44 (m, 2H), 7.36−7.29 (m, 2H), 7.26−7.24 (m, 3H), 6.60 (d, J = 1.7 Hz, 1H), 6.28 (d, J = 1.6 Hz, 1H), 5.97 (s, 1H), 1.16 (s, 6H), 1.11 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 138.4, 136.0, 135.3, 134.3, 133.3, 131.8, 130.6, 128.7, 128.4, 127.5, 121.1, 84.2, 68.9, 24.7, 24.6. $[\alpha]_D^{20}$ = +363.5 (c = 2.2, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{21}H_{24}BBrNaO_4S$ [M + Na]$^+$: 485.0568 and 487.0549; found, 485.0556 and 487.0543. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 5.0 min (minor), 6.5 min (major).

*(R)-2-(3-((4-Bromophenyl)sulfonyl)-3-(4-chlorophenyl)prop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4c**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 75.6 mg, 76% yield, 96% ee, white solid, mp 130−131 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.57 (d, J = 8.7 Hz, 2H), 7.53 (d, J = 8.7 Hz, 2H), 7.31 (d, J = 8.6 Hz, 2H), 7.26 (d, J = 8.6 Hz, 2H), 6.52 (s, 1H), 6.28 (d, J = 1.5 Hz, 1H), 5.09 (s, 1H), 1.15 (s, 6H), 1.11 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 137.9, 135.7, 135.0, 132.1, 131.9, 130.8, 130.7, 129.0, 128.8, 84.4, 71.4, 24.8, 24.5. $[\alpha]_D^{20}$ = +362.2 (c = 1.6, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{21}H_{23}BBrClNaO_4S$ [M + Na]$^+$: 519.0178 and 521.0157; found, 519.0191 and 521.0155. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.1 min (minor), 7.2 min (major).

*(R)-2-(3-((4-Methoxyphenyl)sulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4d**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 75.4 mg, 91% yield, 98% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.55 (d, J = 8.8 Hz, 2H), 7.35−7.32 (m, 2H), 7.27−7.22 (m, 3H), 6.84 (d, J = 9.0 Hz, 2H), 6.54 (s, 1H), 6.25 (d, J = 1.8 Hz, 1H), 5.11 (s, 1H), 3.80 (s, 3H), 1.15 (s, 6H), 1.10 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 163.5, 134.7, 132.8, 131.3, 130.64, 130.57, 128.4, 128.3, 113.8, 84.2, 55.6, 24.8, 24.5. $[\alpha]_D^{20}$ = +229.7 (c = 1.3, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{22}H_{27}BNaO_5S$ [M + Na]$^+$: 437.1569; found, 437.1551. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.2 min (minor), 11.8 min (major).

*(R)-Methyl 3-((1-Phenyl-2-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)allyl)sulfonyl)benzoate (**4e**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 46.0 mg, 52% yield, 96% ee, colorless oil. ^{1}H NMR (400 MHz, $CDCl_3$): δ 8.35 (s, 1H), 8.19 (d, J = 7.8 Hz, 1H), 7.76 (d, J = 7.8 Hz, 1H), 7.46 (t, J = 7.8 Hz, 1H), 7.34−7.32 (m, 2H), 7.27−7.24 (m, 3H), 6.60 (s, 1H), 6.28 (d, J = 1.4 Hz, 1H), 5.15 (s, 1H), 3.91 (s, 3H), 1.12 (s, 6H), 1.08 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 165.5, 139.7, 135.6, 134.2, 133.4, 132.0, 131.0, 130.2, 128.9, 128.8, 128.5, 84.3, 52.7, 24.8, 24.5. $[\alpha]_D^{20}$ = +73.3 (c = 1.0, $CHCl_3$). HRMS (ESI-MS) m/z calcd for $C_{23}H_{27}BNaO_6S$ [M + Na]$^+$: 465.1518; found, 465.1521. HPLC (AD-H, n-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 8.4 min (minor), 10.5 min (major).

*(R)-2-(3-((2-Methoxyphenyl)sulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4f**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 1:1) as eluent, 49.7 mg, 60% yield, 98% ee, white solid, mp 183−184 °C. ^{1}H NMR (400 MHz, $CDCl_3$): δ 7.70−7.68 (m, 1H), 7.48−7.42 (m, 3H), 7.24−7.22 (m, 3H), 6.96−6.89 (m, 2H), 6.51 (s, 1H), 6.79 (d, J = 2.0 Hz, 1H), 5.78 (s, 1H), 4.00 (s, 3H), 1.14 (s, 6H), 1.07 (s, 6H). ^{13}C{^{1}H} NMR (101 MHz, $CDCl_3$): δ 157.4, 135.4, 135.2, 132.3, 131.6, 130.7, 128.4, 128.3, 127.3, 120.5, 112.0, 84.0, 69.5, 56.3, 24.9,

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24.5. $[\alpha]_D^{20}$ = +406.5 (*c* = 1.0, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{22}H_{27}BNaO_5S$ $[M + Na]^+$: 437.1569; found, 437.1569. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.8 min (minor), 13.3 min (major).

*(R)-2-(3-((1,3-Dihydroisobenzofuran-5-yl)sulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4g**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 1:1) as eluent, 69.1 mg, 81% yield, 98% ee, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.43−7.40 (m, 2H), 7.36−7.34 (m, 2H), 7.27−7.23 (m, 3H), 6.72 (d, *J* = 8.2 Hz, 1H), 6.56 (s, 1H), 6.27 (d, *J* = 1.5 Hz, 1H), 5.10 (s, 1H), 4.63 (t, *J* = 8.8 Hz, 2H), 3.24−3.08 (m, 2H), 1.16 (s, 6H), 1.11 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 164.4, 134.9, 132.9, 131.0, 130.7, 130.6, 128.4, 128.3, 127.8, 126.4, 109.2, 84.2, 72.4, 29.0, 24.8, 24.5. $[\alpha]_D^{20}$ = +216.5 (*c* = 1.0, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{23}H_{27}BNaO_5S$ $[M + Na]^+$: 449.1569; found, 449.1558. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 7.2 min (minor), 9.8 min (major).

*(R)-4,4,5,5-Tetramethyl-2-(3-phenyl-3-(thiophen-2-ylsulfonyl)-prop-1-en-2-yl)-1,3,2-dioxaborolane (**4h**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 1:1) as eluent, 55.4 mg, 71% yield, 98% ee, white solid, mp 92−93 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.59 (d, *J* = 4.8 Hz, 1H), 7.40−7.39 (m, 3H), 7.30−7.28 (m, 3H), 7.00 (t, *J* = 4.2 Hz, 1H), 6.61 (s, 1H), 6.30 (d, *J* = 2.0 Hz, 1H), 5.24 (s, 1H), 1.16 (s, 6H), 1.12 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 139.9, 135.2, 135.1, 134.1, 132.5, 130.5, 128.7, 128.5, 127.4, 84.3, 73.3, 24.8, 24.6. $[\alpha]_D^{20}$ = +174.8 (*c* = 2.2, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{19}H_{23}BNaO_4S_2$ $[M + Na]^+$: 413.1027; found, 413.1023. HPLC (AD-H, *n*-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 8.5 min (minor), 13.3 min (major).

*(R)-2-(3-(Butylsulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4i**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 53.9 mg, 74% yield, 99% ee, pale yellow oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.54−7.52 (m, 2H), 7.39−7.33 (m, 3H), 6.49 (s, 1H), 6.30 (d, *J* = 2.0 Hz, 1H), 5.11 (s, 1H), 2.87 (t, *J* = 8.0 Hz, 2H), 1.84−1.72 (m, 2H), 1.42−1.33 (m, 2H), 1.23 (s, 6H), 1.20 (s, 6H), 0.89 (t, *J* = 7.4 Hz, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 135.4, 132.3, 130.3, 128.7, 84.4, 69.0, 51.7, 24.8, 24.7, 23.8, 21.8, 13.6. $[\alpha]_D^{20}$ = +46.7 (*c* = 1.8, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{19}H_{29}BNaO_4S$ $[M + Na]^+$: 387.1775; found, 387.1772. HPLC (AD-H, *n*-hexane/isopropanol = 95/5, flow rate = 1.0 mL/min, l = 220 nm) tR = 11.5 min (minor), 12.3 min (major).

*(R)-2-(3-(Benzylsulfonyl)-3-phenylprop-1-en-2-yl)-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (**4j**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 44.6 mg, 56% yield, 97% ee, pale yellow oil. ^1H NMR (400 MHz, $CDCl_3$); δ 7.49−7.47 (m, 2H), 7.37−7.32 (m, 8H), 6.52 (s, 1H), 6.33 (d, *J* = 1.8 Hz, 1H), 5.00 (s, 1H), 4.25 (d, *J* = 13.7 Hz, 1H), 4.13 (d, *J* = 13.8 Hz, 1H), 1.21 (s, 6H), 1.18 (s, 6H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 135.1, 131.5, 131.0, 130.7, 128.9, 128.8, 128.7, 128.0, 84.3, 68.0, 58.0, 24.8, 21.7. $[\alpha]_D^{20}$ = +67.6 (*c* = 1.0, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{22}H_{27}BNaO_4S$ $[M + Na]^+$: 421.1619; found, 421.1614. HPLC (AD-H, *n*-hexane/isopropanol = 70/30, flow rate = 1.0 mL/min, l = 220 nm) tR = 6.0 min (minor), 6.4 min (major).

*Methyl (Z)-2-Phenyl-3-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)but-2-enoate (**6**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 38.8 mg, 64% yield, colorless oil. ^1H NMR (400 MHz, $CDCl_3$): δ 7.38−7.28 (m, 3H), 7.15−7.12 (m, 2H), 3.74 (s, 3H), 1.78 (s, 3H), 1.39 (s, 12H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 170.8, 137.3, 135.2, 129.7, 128.2, 127.5, 83.6, 53.1, 25.0, 18.0. HRMS (ESI-MS) *m*/*z* calcd for $C_{17}H_{23}BNaO_4$ $[M + Na]^+$: 325.1585; found, 325.1576.

General Procedure for the Halogenation of Compound 2a. A mixture of **2a** (39.8 mg, 0.1 mmol), CuX_2 (0.3 mmol, 3.0 equiv, X = Cl or Br) and MeOH/H_2O (1 mL/1 mL) was heat to 80 °C under air atmosphere in a sealed tube. The reaction was stirred at 80 °C in an oil bath for 6 h. The reaction was cooled to rt and quenched with brine (1 mL). The mixture was extracted with EtOAc (5 mL × 3). The combined organic phase was dried over anhydrous Na_2SO_4, filtered, and concentrated. The resultant residue was purified by flash chromatography on silica gel to afford the target compound.

*(R)-1-((2-Chloro-1-phenylallyl)sulfonyl)-4-methylbenzene (**2aa**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 29.2 mg, 95% yield, >99% ee, white solid, mp 126−127 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.54 (d, *J* = 8.3 Hz, 2H), 7.37−7.29 (m, 5H), 7.23 (d, *J* = 8.1 Hz, 2H), 6.14 (d, *J* = 2.2 Hz, 1H), 5.68 (d, *J* = 2.3 Hz, 1H), 4.88 (s, 1H), 2.41 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 145.2, 134.8, 133.3, 130.3, 130.2, 129.6, 129.5, 129.3, 128.7, 119.4, 77.1, 21.8. $[\alpha]_D^{20}$ = +652.7 (*c* = 1.0, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{16}H_{15}ClNaO_2S$ $[M + Na]^+$: 329.0373 and 331.0347; found, 329.0364 and 331.0337. HPLC (IC−H, *n*-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 25.1 min (major).

*(R)-1-((2-Bromo-1-phenylallyl)sulfonyl)-4-methylbenzene (**2ab**).* Purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent, 26.7 mg, 76% yield, >99% ee, white solid, mp 106−107 °C. ^1H NMR (400 MHz, $CDCl_3$): δ 7.52 (d, *J* = 8.2 Hz, 2H), 7.38−7.37 (m, 1H), 7.31−7.30 (m, 4H), 7.22 (d, *J* = 8.1 Hz, 2H), 6.70 (d, *J* = 2.6 Hz, 1H), 5.97 (d, *J* = 2.7 Hz, 1H), 4.96 (s, 1H), 2.41 (s, 3H). $^{13}C\{^1H\}$ NMR (101 MHz, $CDCl_3$): δ 145.2, 134.9, 130.34, 130.27, 129.64, 129.57, 129.2, 128.7, 123.5, 122.8, 78.5, 21.8. $[\alpha]_D^{20}$ = +544.3 (*c* = 1.9, $CHCl_3$). HRMS (ESI-MS) *m*/*z* calcd for $C_{16}H_{15}BrNaO_2S$ $[M + Na]^+$: 372.9868 and 374.9848; found, 372.9863 and 374.9843. HPLC (IC−H, *n*-hexane/isopropanol = 90/10, flow rate = 1.0 mL/min, l = 220 nm) tR = 24.0 min (major).

Procedure for the Oxidation of Compound 2a. A mixture of **2a** (39.8 mg, 0.1 mmol, 1.0 equiv), $NaBO_3{\cdot}4H_2O$ (77.0 mg, 0.5 mmol, 5.0 equiv) in THF/H_2O (1 mL/1 mL) was stirred at room temperature for 1 h. The reaction mixture was dissolved in EtOAc (30 mL) and washed with brine (10 mL × 2). The organic phase was dried over anhydrous Na_2SO_4, filtered, and concentrated. The resultant residue was purified by column chromatography on silica gel with petroleum ether/EtOAc (v/v = 3:1) as eluent to afford the compound **7** (21.1 mg, 73% yield).

■ ASSOCIATED CONTENT

Data Availability Statement

The data underlying this study are available in the published article and its Supporting Information.

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/acs.joc.4c02849.

Spectra of new compounds, including ^1H and ^{13}C NMR. X-ray crystallographic analysis of compound **4c**; CCDC number for **4c** is 2361361 (PDF)

Accession Codes

Deposition Number 2361361 contains the supplementary crystallographic data for this paper. These data can be obtained free of charge via the joint Cambridge Crystallographic Data Centre (CCDC) and Fachinformationszentrum Karlsruhe Access Structures service.

■ AUTHOR INFORMATION

Corresponding Authors

Tongyu Xu − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an 710127, P. R. China;* orcid.org/0000-0001-5337-2949; Email: xutongyu@nwu.edu.cn

Fanlong Zeng − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an*

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710127, P. R. China; orcid.org/0000-0002-4293-5133; Email: fzeng@nwu.edu.cn

Authors

Peidong Sun − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an 710127, P. R. China*

Xiaomei Kong − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an 710127, P. R. China*

Jingwen Huang − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an 710127, P. R. China*

Xuedi Ma − *Key Laboratory of Synthetic and Natural Functional Molecule of the Ministry of Education, College of Chemistry & Materials Science, Northwest University, Xi'an 710127, P. R. China*

Complete contact information is available at: https://pubs.acs.org/10.1021/acs.joc.4c02849

Author Contributions

†P.S. and X.K. contributed equally. The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

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